A Web Based Framework For Vehicle Parking Reservation And Space Management

Project Phase - I Report submitted to Osmania University, Hyderabad, in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering

In

Information Technology

By

Mr. Abdul Rahman Baig (160923737301)

Under the Guidance of

Mr. Syed Juber

(Assistant Professor)

DEPARTMENT OF INFORMATION TECHNOLOGY

LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY

(UGC Autonomous)

(Affiliated to Osmania University) Himayathsagar, Hyderabad,500091

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Himayathsagar, Hyderabad, 500091
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This is to certify that the Project Phase-I report entitled **A Web Based Framework For Vehicle Parking Reservation And Space Management** being submitted by **Abdul Rahman Baig (160923737301)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology** is a record of Bonafide work carried out by them.

Internal Project Guide		
Mr. Syed Juber	**External Examiner**	**Dr. K. Nagi Reddy**
(Assistant Professor)		**H.O.D, I.T**



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DECLARATION BY THE CANDIDATES

We, **Abdul Rahman Baig (160923737301)** hereby declare that the Project Phase-I report entitled **A Web Based Framework For Vehicle Parking Reservation And Space Management** is carried out under the guidance of **Mr. Syed Juber** Assistant Professor, Department of Information Technology, **Lords Institute of Engineering and Technology**, **Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology.**

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We are very pleased to present this thesis of our project. This period of our student life has been truly rewarding a number of people were of immense help to us during the course of our research and the preparation of our thesis.

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Abdul Rahman Baig (160923737301)

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PEO2	Exhibit professionalism, engage in lifelong learning and adopt to changing industry and societal needs.
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2.	**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
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4.	**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
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PSO2	**Problem-Solving Skills:** Deliver quality service to IT industry and provide solutions to real-life problems applying modern computer languages, software methods and tools.

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Course Outcomes: Project Phase- I

After completion of the project, the student will be able to

S.No	Description	Blooms Taxonomy Level
PP.I.CO1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	BTL3
PP.I.CO2	Evaluate different solutions based on economic and technical feasibility	BTL4
PP.I.CO3	Effectively plan a project and confidently perform all aspects of project management.	BTL5
PP.I.CO4	Demonstrate effective written and oral communication skills	BTL4
PP.I.CO5	Prepare the documentation report and perform the presentation of the project work	BTL3

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Course Articulation Matrix:

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2
PP.I.CO1	3	2	2	-	2	2	-	-	3	-	-	3	2
PP.I.CO2	3	3	2	3	2	-	2	-	3	-	-	3	2
PP.I.CO3	3	3	3	-	3	3	1	2	2	3	2	2	3
PP.I.CO4	3	2	2	3	3	3	3	2	-	-	2	2	3
PP.I.CO5	3	3	3	3	3	3	3	2	3	3	-	3	2
AVG	**3**	**2.6**	**2.5**	**3**	**2.6**	**2.7**	**2.2**	**2**	**2.9**	**3**	**2**	**2.6**	**2.4**

Level:

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	✓
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	✓	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES	✓	17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

ABSTRACT

Finding an available parking space in busy commercial areas, educational institutions, hospitals, offices, residential communities, and transport hubs can be inconvenient. Traditional parking operations often depend on drivers searching for a space on arrival, manual registers, paper permits, or disconnected payment and access processes. These methods may lead to unnecessary circulation, queues, uncertain availability, inefficient use of parking capacity, and limited visibility for facility managers.

The project, A Web-Based Framework for Vehicle Parking Reservation and Space Management, proposes a centralized web application through which users can discover parking facilities, review space availability, reserve a suitable space for a selected time window, and manage their bookings. The system maintains information about parking locations, space identifiers, vehicle categories, opening hours, reservation windows, rates, and booking status. It provides an administrative interface for facility managers to configure spaces, monitor reservations, handle cancellations, and review operational summaries. A core design requirement is preventing conflicting reservations. The framework validates a requested time window against existing confirmed bookings and applies a consistent reservation rule before confirming a space. Database constraints, transactional operations, and suitable locking or conflict checks are used to reduce the risk that two users reserve the same space for overlapping periods. Booking statuses such as pending, confirmed, cancelled, checked-in, completed, and no-show support a traceable reservation lifecycle. Payment integration, QR-based check-in, and sensor-driven occupancy are optional extensions unless implemented and tested as part of the prototype.

It incorporates authentication, role-based access, input validation, audit logging, and protection of user and vehicle details. The expected outcome is a practical and extensible platform that improves reservation convenience, makes parking capacity easier to manage, and provides a foundation for data-driven facility operations. Effectiveness should be evaluated through functional tests, booking-conflict tests, usability feedback, and response-time measurements under a defined workload.

Keywords: Vehicle Parking, Web Application, Reservation System, Availability Tracking, Rest API, Relational Database, Role-Based Access Control.

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CHAPTER-1
INTRODUCTION

1.1 Background and Motivation

Parking is an essential part of urban mobility and the operation of facilities that receive visitors. As vehicle ownership and private transport use increase, drivers may spend time searching for a suitable space, especially during peak hours. The problem is not always a shortage of physical spaces; it can also arise from poor information about available capacity, uneven demand, inefficient allocation, and limited coordination between users and parking operators.

In many small and medium facilities, entry and exit are managed by attendants who inspect spaces visually or maintain paper records. Some facilities use ticketing machines or payment applications, but these systems may not offer advance reservations or a consolidated view of capacity. Drivers often discover that a preferred area is full only after arriving. Facility operators may also have difficulty identifying peak periods, cancelled bookings, no-shows, or spaces that remain underused.

A web-based parking reservation framework can connect drivers and parking administrators through a shared source of information. Users can search for facilities, select a date and time, review pricing and restrictions, and reserve an eligible space. Administrators can update facility details, define space categories, manage reservations, and review utilization summaries. A browser-based solution can be accessed from computers and mobile devices without requiring a dedicated application.

The proposed system manages reservations and space information rather than claiming real-time physical occupancy automatically. Unless sensors, cameras, gates, or attendant check-in procedures are integrated, the application knows the reservation state but cannot independently guarantee that a vehicle is physically occupying a space. The design therefore distinguishes a reserved space from a space verified as physically vacant.[1]

1.2 Problem Statement

Drivers frequently lack reliable information about parking availability and may be unable to reserve a suitable space before reaching a destination. Manual or disconnected processes can create queues, duplicate allocations, inconsistent records, and

administrative overhead. Facility managers may have limited tools to coordinate space inventory, booking windows, cancellations, and utilization reporting.

The problem addressed by this project is to design a web-based framework that maintains parking facilities and spaces, supports time-based reservations, checks availability before confirming bookings, prevents overlapping reservations for the same space, and provides clear booking and management interfaces for users and administrators.[1]

1.3 Aim and Objectives

The main aim is to develop a secure, user-friendly web application for parking discovery, reservation, and parking-space administration.

- Allow users to browse and search parking facilities using location, date, time, vehicle type, and supported filters.
- Display facility information such as address, operating hours, available space categories, pricing, restrictions, and amenities.
- Allow authenticated users to create, view, and cancel reservations according to policy.
- Maintain parking facilities, spaces, rates, operating periods, and reservation records in a consistent database.
- Validate requested reservation windows against existing bookings and prevent overlapping confirmed reservations.
- Provide administrative functions for managing facilities, spaces, rates, availability, and booking status.
- Support a reservation lifecycle including confirmation, cancellation, check-in, completion, and no-show where applicable.
- Apply authentication, role-based authorization, input validation, and secure handling of user and vehicle data.
- Generate operational summaries such as reservations by date, cancellation counts, and utilization estimates.
- Design for future payment gateways, QR check-in, notifications, and sensor integration.

1.4 Scope of the Project

The project scope includes a responsive web interface, backend services, a relational database, user and administrator roles, parking facility and space management, availability search, time-window reservation, booking history, cancellation rules, and administrative reports. The system may include simulated payment status or a payment-provider sandbox if selected for implementation. Real payments should not be represented as complete unless a payment gateway is integrated and tested.

The initial version focuses on defined facilities and structured booking rules. It may support cars and two-wheelers, different space types, hourly or fixed-duration slots, and configurable rates. Location search may use stored addresses or map integration depending on implementation. The system can be demonstrated with seeded sample facilities and users.

Automatic occupancy detection, license-plate recognition, barrier control, advanced route optimization, dynamic pricing, and city-wide parking aggregation are outside the mandatory core prototype. These functions require additional hardware, integrations, privacy safeguards, and operational testing. The application should clearly distinguish a reserved space from a space verified as physically vacant.

1.5 Significance and Expected Outcomes

The framework is expected to improve users' ability to plan parking before arrival and reduce uncertainty about reservation status. A centralized inventory can help facility operators maintain accurate records, apply consistent booking rules, and review demand patterns. Conflict prevention can reduce duplicate allocations, while clear cancellation and check-in states improve operational traceability.

Expected outcomes include user registration and login, facility search, availability display, reservation creation and cancellation, administrator management screens, booking status tracking, conflict-prevention logic, operational reports, and test evidence. The project can provide a foundation for future smart-parking features, provided the core data model and APIs are designed with extensibility in mind.[2]

1.6 Organization of the Report

Chapter 1 introduces the parking problem, objectives, scope, and expected outcomes. Chapter 2 surveys existing parking approaches, web reservation systems, smart-parking

concepts, scheduling, security, and related technologies. Chapter 3 analyzes existing practices and specifies the proposed system's requirements and feasibility. Chapter 4 describes architecture, modules, database, reservation logic, APIs, and security design. Chapter 5 explains input/output design, implementation, validation, and testing. Chapter 6 documents the software environment, deployment considerations, conclusion, and future enhancements.[3]

CHAPTER-2
LITERATURE SURVEY

2.1 Parking Management Approaches

Parking management has evolved from manual attendants and paper tickets to automated ticketing, access-control equipment, online payments, and digital reservation platforms. Manual approaches are inexpensive and can be sufficient for small lots, but accuracy depends on staff procedures and timely updates. Ticketing systems improve entry and exit records but do not necessarily provide advance reservation or a unified view of availability across facilities.

Digital parking management systems typically maintain an inventory of spaces, vehicle categories, rates, opening hours, and transactions. Their value depends on whether the recorded state reflects the physical facility. A system based only on reservations can accurately report bookings while still lacking evidence of actual occupancy. Effective designs distinguish planned allocation, check-in records, and sensor-confirmed occupancy.[2]

2.2 Web-Based Reservation Systems

Web-based reservation systems use a browser interface and server-side services to let users search resources and reserve them for a time window. Similar patterns appear in hotel booking, appointment scheduling, equipment rental, and event ticketing. Common capabilities include availability search, user accounts, booking confirmation, cancellation, reminders, payment status, and administrative management.

Parking reservations introduce specific constraints. A space is a reusable resource rather than a one-time ticket, and two bookings may use the same space if their time windows do not overlap and adequate turnover time is respected. The system must account for operating hours, vehicle size or type, rates, grace periods, early arrivals, overstays, cancellations, and no-shows. These rules should be explicit rather than embedded in unrelated user-interface code.

A web framework can centralize business rules in backend services and expose them through documented APIs. The frontend can display available options authoritative when checking availability and confirming reservations. This matters because client-side availability can become stale when several users search or book concurrently[1]

2.3 Smart Parking and Availability Detection

Smart parking systems combine software with technologies such as ultrasonic sensors, magnetic sensors, cameras, entry counters, connected parking meters, or gate systems. Such devices can estimate or report whether a space is occupied. Sensor-driven information can reduce uncertainty, but it introduces hardware costs, connectivity requirements, calibration needs, and potential errors caused by sensor failure or environmental conditions.

A reservation platform can be useful without sensors if it reports reservation availability rather than claiming verified physical vacancy. Attendants can update check-in and checkout events, or users can confirm arrival through a QR code. Later, sensor or gate integrations can update occupancy records and compare physical occupancy with reservations. The data model should preserve the source and timestamp of each status update so conflicting signals can be investigated.

Smart parking literature also emphasizes timely information. A displayed count may be inaccurate if updates are delayed or drivers occupy spaces without registering. The proposed project treats live occupancy as an optional integration and establishes a core reservation workflow that can be tested independently.[3]

2.4 Reservation Scheduling and Conflict Prevention

Resource scheduling requires checking whether a requested interval conflicts with existing allocations. For half-open intervals, a booking from start A to end A overlaps a booking from start B to end B when A is earlier than end B and B is earlier than end A. This convention allows a reservation ending at 10:00 and another starting at 10:00 to be adjacent if the facility's turnover policy permits it.

Checking availability only in the browser is insufficient. Two users may request the same space at nearly the same time after both have seen it as available. The backend must perform the conflict check within a transaction and use a concurrency-control approach appropriate to the database. Depending on the database, this may involve row-level locking, exclusion constraints, serializable transactions, or another reliable mechanism. A rejected conflicting booking should produce a clear message and alternative options.

Booking lifecycle rules affect capacity. A cancelled booking should no longer block the interval, while a confirmed booking should. A pending reservation may temporarily

hold capacity only if the system defines an expiry policy. No-show handling, check-in grace periods, and overstays should be configurable and documented.

2.5 Usability, Security and Privacy

A useful booking interface minimizes the steps needed to find a facility and understand its terms. Important information includes location, entry instructions, opening hours, total price, cancellation rules, space type, height restrictions, accessibility, and whether availability is based on reservation records or verified occupancy. Mobile responsiveness is important because users often search while travelling.

Security controls commonly include secure authentication, role-based access, server-side validation, protection against injection and cross-site scripting, secure session or token management, rate limiting, and audit logging for administrative changes. Users should only access their own booking details, while facility administrators should be limited to the facilities they manage where such separation is required.

Vehicle registration numbers, contact details, location history, and payment references can be sensitive. A system should collect only necessary information, restrict access, avoid exposing details in URLs or logs, define retention periods, and use HTTPS in deployment. Payment card data should be handled by a compliant payment provider rather than stored directly by the prototype.[4]

2.6 Related Technologies

A typical web solution separates a frontend, backend API, and persistence layer. Frontend frameworks provide reusable components for search forms, availability cards, reservation details, and administrative tables. Backend frameworks support authentication, validation, business services, and transactional database access. Relational databases suit entities such as users, facilities, spaces, reservations, rates, and payment records because these entities have defined relationships and consistency requirements.

REST APIs are a common integration style for browser clients and backend services. Endpoints can represent facilities, spaces, availability queries, reservations, user profiles, and administrative operations. API validation should enforce the same rules as the user interface because requests can be submitted outside the normal browser workflow. Database migrations, version control, automated tests, and structured

logging support maintainability.

Map services can support location-based discovery, while email or SMS services can send booking confirmations and reminders. Payment gateways can process payments and return transaction references. QR codes may encode a short-lived booking reference for check-in. Each external service adds dependency, cost, failure handling, and data-protection requirements; integrations should be added incrementally.[4]

2.7 Research Gap and Project Positioning

Existing parking solutions range from manual operations to sophisticated smart-city systems. Some focus on occupancy detection, others on payments or access control, and some on advance booking. For a student project, a practical challenge is to combine a usable reservation flow with consistent space inventory, conflict-safe time-based allocation, clear administrative controls, and secure data handling without requiring specialized hardware.

The proposed framework focuses on this integration. Its core contribution is a coherent web-based workflow that treats availability as a time-dependent property, validates reservations on the server, records booking lifecycle changes, and gives users and administrators distinct interfaces. The project does not claim to invent a new scheduling algorithm or provide city-wide real-time occupancy. It provides an extensible implementation of established web, database, and reservation-management principles.

Evaluation should consider more than whether a booking can be created. It should test overlapping requests, cancellation and rebooking, operating-hour restrictions, role permissions, stale availability, failure recovery, and user understanding of reservation status.[5]

CHAPTER-3
SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Existing System

In a basic parking facility, drivers arrive and search for an open space. An attendant may issue a paper ticket, record a vehicle number, collect payment, or direct the driver to a section. Some facilities use separate ticketing or payment software, but booking information may not be available before arrival. A user who needs parking at a particular time may have to call the facility or rely on informal information.

This approach can work when demand is low and staff can observe the lot. During peak periods, however, it may produce queues, uncertainty, repeated circulation, and inconsistent records. Manual records also make it harder to analyze utilization, no-shows, cancellation patterns, and demand by time of day. The limitations vary by facility; the proposed system improves coordination rather than assuming every manual facility experiences every problem.[5]

3.2 Limitations of Existing Approaches

- Availability may not be visible to drivers before arrival.
- Manual updates can be delayed or inconsistent across staff members.
- Users may have no standard way to reserve a space for a future time window.
- Duplicate or overlapping allocations can occur if booking records are not checked centrally.
- Cancellation, no-show, and overstay rules may be applied inconsistently.
- Facility managers may lack consolidated reports about bookings and utilization.
- Payment, check-in, and reservation data may be maintained in separate systems.
- Users may have limited visibility into rates, restrictions, and cancellation conditions.

3.3 Proposed System

The proposed framework maintains a central catalogue of parking facilities and their spaces. Users can browse facilities, filter by supported criteria, select a time window, and request a reservation. The backend checks that the facility is open, the space

supports the requested vehicle category, the interval is valid, and no active reservation conflicts with the request. If all checks pass, the system confirms the booking and returns a unique booking reference.

Users can review booking details and cancel within the applicable policy. Administrators can add or update facilities, manage spaces and rates, view reservations, and update operational states such as check-in or completion. A report module summarizes booking counts, cancellations, and estimated utilization. If payment is included, reservation and payment states are tracked separately so that a failed payment does not incorrectly appear as a completed transaction.

The backend remains the authoritative source for reservation decisions. The frontend may show availability as a snapshot, but every booking request is revalidated on the server. Conflicting concurrent requests should result in at most one confirmed reservation for a given space and overlapping interval.[4]

3.4 Functional Requirements

FR-01 User registration and login: The system shall allow users to register, authenticate, and securely access their account.

FR-02 Role management: The system shall distinguish regular users from administrators and enforce permissions on the server.

FR-03 Facility browsing: Users shall browse facilities and view address, operating hours, rates, restrictions, and supported space categories.

FR-04 Search and filtering: Users shall search by supported criteria such as date, time, vehicle category, and location.

FR-05 Availability query: The system shall calculate eligible spaces for a requested time window using active reservation records and facility rules.

FR-06 Reservation creation: Authenticated users shall request a space for a valid interval and receive a unique reference when confirmed.

FR-07 Conflict prevention: The backend shall prevent overlapping confirmed reservations for the same space.

FR-08 Booking history: Users shall view their own current and past reservations.

FR-09 Cancellation: Users shall cancel eligible reservations according to configured policy.

FR-10 Administration: Authorized administrators shall manage facilities, spaces,

rates, operating hours, and restrictions.

FR-11 Booking status: Authorized staff shall record permitted status transitions such as check-in and completion.

FR-12 Notifications: The system should provide booking confirmations and cancellation notifications through the selected channel.

FR-13 Reporting: Administrators shall view booking, cancellation, and estimated utilization summaries.

FR-14 Audit trail: The system should record significant administrative and booking status changes.

FR-15 Data export: Authorized administrators may export summarized reservation data.

FR-16 Integration readiness: The architecture should allow future payment, map, QR check-in, or sensor integrations.

3.5 Non-Functional Requirements

Usability: The interface should be clear, responsive, and usable on desktop and mobile browsers.

Performance: Common searches and booking requests should respond within a defined target under a documented workload.

Consistency: Reservation confirmation must preserve the no-overlapping-bookings rule under concurrent requests.

Security: Authentication, authorization, validation, secure transport, and safe error handling must protect accounts and booking data.

Privacy: The application should minimize personal data, restrict access, and define retention and deletion practices.

Reliability: Errors in notifications or optional integrations should not silently corrupt reservation state.

Maintainability: Frontend, API, reservation services, persistence, and integrations should have clear responsibilities.

Scalability: The design should support additional facilities and concurrent users through indexed queries and controlled scaling.

Availability: Deployment should define monitoring, backup, and recovery expectations appropriate to the prototype.

Portability: The application should run in a documented development environment and deploy to a supported platform.

Auditability: Important booking and administrative changes should be traceable to a user or system action.

Accessibility: Forms, status messages, navigation, and controls should follow appropriate web accessibility practices.[6]

3.6 User Roles and Use Cases

User Role

A registered driver can search facilities, inspect available options, create reservations, review booking history, and cancel eligible bookings. The user must not be able to view or modify another user's private reservation data.

Administrator Role

An administrator can maintain facility information, space inventory, rates, operating hours, restrictions, and booking status within the assigned scope. Administrative actions should be validated and logged. Multi-operator deployments should scope permissions so one operator cannot alter another operator's facilities.

Use Case 1: Reserve a parking space

1. The user opens the application and selects a facility or search criteria.
2. The system displays facilities and eligible space categories for the requested date and time.
3. The user selects an option and reviews the total price and booking terms.
4. The authenticated user submits the reservation request.
5. The backend validates interval, facility hours, eligibility, and conflicts within a safe transaction.
6. The system confirms the reservation or returns a conflict message with alternative options.
7. The user views the booking reference and reservation details.

Use Case 2: Manage parking inventory

8. The administrator signs in and opens facility management.
9. The administrator creates or edits a facility, space, rate, or operating-hour record.
10. The system validates required fields and checks for inconsistent configurations.
11. The change is saved and recorded in the audit trail.
12. The updated inventory is reflected in subsequent availability queries.

3.7 Feasibility Study

Technical Feasibility

The project is technically feasible using established web frameworks, REST APIs, and a relational database. Reservation logic, role-based access, and administrative interfaces are standard application-development tasks. A prototype can be developed and tested locally without specialized parking hardware. Optional map, payment, notification, or sensor integrations can be added when credentials and clear requirements are available.

Economic Feasibility

The core application can be built with open-source development tools, limiting licensing expenditure. Costs may arise from hosting, database services, map requests, SMS or email delivery, payment-provider fees, and monitoring. Costs can be controlled by using a local or low-cost test environment and delaying paid integrations until the core workflow is stable.

Operational Feasibility

The workflow of searching, selecting, reserving, and managing a booking is familiar to users. Facility administrators will require a clear process for keeping inventory and operating hours accurate. The system's value depends on trustworthy data and staff adherence to check-in, cancellation, and occupancy procedures.

Schedule Feasibility

Implementation can proceed in phases: requirements and data model; authentication and facility inventory; availability and reservation logic; administrative functions; reporting and notifications; then testing, documentation, and deployment. Payment, QR entry, or sensor integrations can be treated as optional extensions.[5]

CHAPTER-4
SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The architecture consists of a presentation layer, backend API, business-service layer, relational database, and optional integration adapters. The presentation layer contains public facility pages, search results, reservation forms, booking history, and administrator screens. The backend API authenticates requests, validates inputs, applies access controls, and coordinates services. The service layer contains reservation, availability, facility management, pricing, notification, and reporting logic. The database stores users, facilities, spaces, rates, reservations, and audit events.

A modular monolith is appropriate for the first implementation because the main workflows share transactional data and can be developed and tested with less operational complexity than a distributed microservice architecture. Modules can be separated internally so a future deployment can extract high-volume or independently managed services if evidence justifies it.

The database is the authoritative source for confirmed bookings. Availability results shown to a user are advisory until the reservation transaction succeeds. The booking endpoint must repeat relevant checks to account for changes made by other users after the availability query.[6]

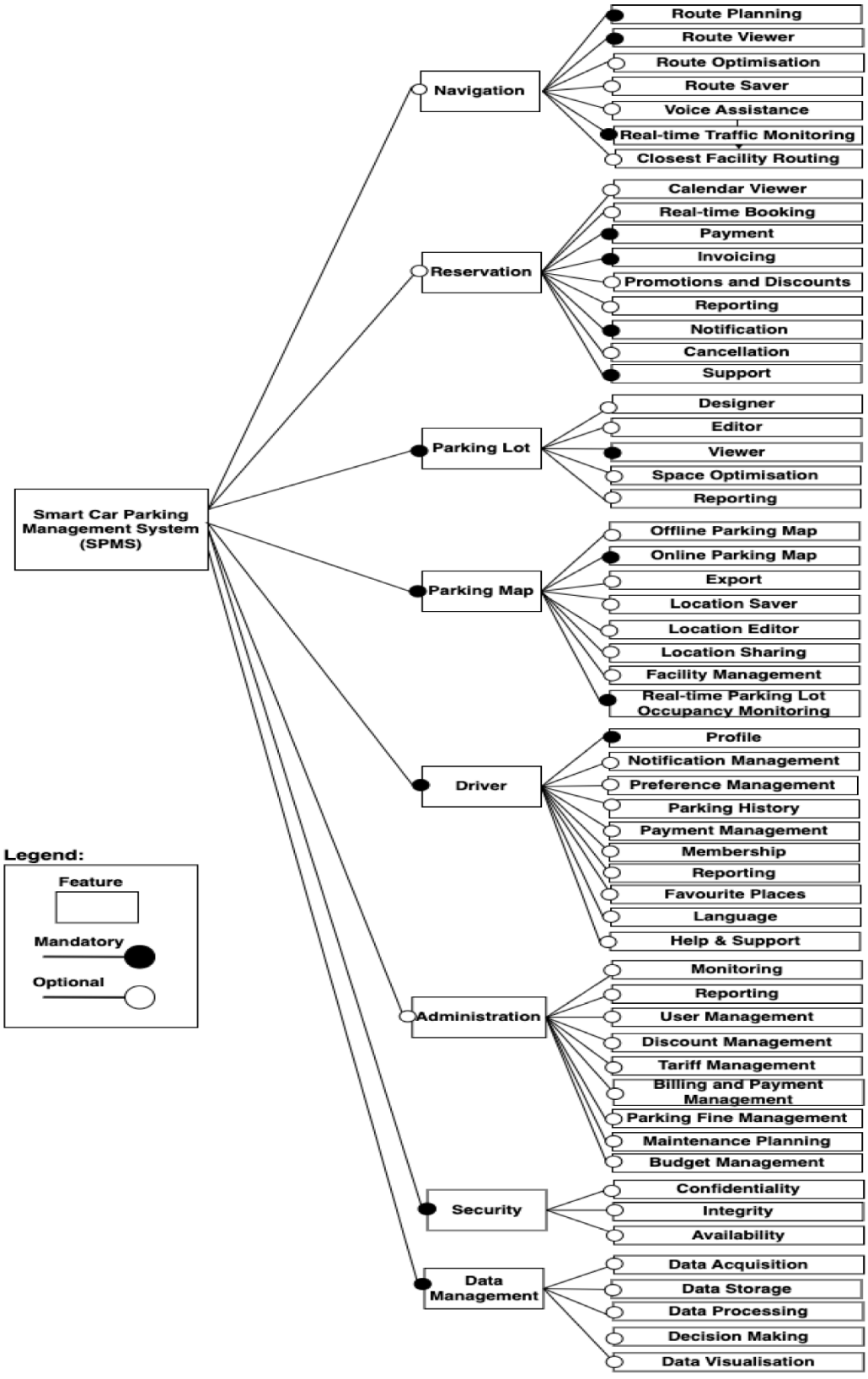


Fig 4.1 System Architecture

4.2 System Workflow

13. User browses or searches parking facilities.
14. Frontend sends search criteria including vehicle category and requested time window.
15. Backend validates criteria and queries facilities, spaces, operating hours, and active reservations.
16. Frontend displays eligible options, prices, restrictions, and applicable time window.
17. Authenticated user submits a reservation request.
18. Reservation service validates the request and performs conflict prevention within a transaction.
19. On success, the service stores a confirmed reservation and returns a booking reference; otherwise it returns a conflict or validation response.
20. The system sends a confirmation through the selected notification mechanism after the reservation is committed.
21. User or authorized staff records cancellation, check-in, completion, or no-show according to allowed transitions.
22. Reporting module aggregates reservation data for administrative review.[7]

4.3 Module Design

User and Authentication Module: Manages registration, login, account details, password handling or identity-provider integration, and session/token lifecycle.

Facility Catalogue Module: Maintains facilities, addresses, operating hours, amenities, supported vehicle categories, and facility status.

Space Inventory Module: Stores individual spaces or capacity units, categories, identifiers, restrictions, and active status.

Search and Availability Module: Filters eligible facilities and calculates availability for a requested interval.

Reservation Module: Creates, retrieves, and cancels bookings while enforcing conflict and lifecycle rules.

Pricing Module: Calculates charges from configured rates, duration, vehicle category, discounts, and applicable fees.

Administration Module: Provides authorized screens for facility, space, rate, support,

and booking operations.

Notification Module: Sends confirmation, cancellation, reminder, or operational messages through configured providers.

Reporting Module: Produces summaries of bookings, cancellations, estimated utilization, and no-show counts.

Audit and Integration Module: Records important state changes and isolates optional payment, map, QR, and sensor integrations.

4.4 Database Design

A relational data model is suitable because reservations depend on users, spaces, facilities, and time intervals. Primary keys identify records, while foreign keys maintain relationships. Indexes should support facility searches and conflict checks. Exact types, constraints, and indexes depend on the selected database engine.

Core Entities

User: user_id (PK), name, email, password_hash or identity reference, optional phone, role, account_status, created_at.

Facility: facility_id (PK), name, address, optional latitude/longitude, opening_hours, description, facility_status.

ParkingSpace: space_id (PK), facility_id (FK), space_code, vehicle_category, space_type, restrictions, active_status.

Rate: rate_id (PK), facility or space-type reference, pricing_unit, amount, effective_from, effective_to, currency.

Reservation: reservation_id (PK), booking_reference, user_id (FK), space_id (FK), start_time, end_time, status, quoted_amount, created_at, cancelled_at.

PaymentRecord (optional): payment_id (PK), reservation_id (FK), provider_reference, amount, currency, payment_status, updated_at.

AuditEvent: event_id (PK), actor_user_id where applicable, entity_type, entity_id, action, timestamp, safe change summary.

OccupancyEvent (optional): event_id (PK), space_id (FK), state, source, observed_at, confidence or verification details where supported.

Reservation start and end timestamps should follow a clearly defined time-zone policy. The system should reject intervals where end time is not later than start time. If a facility uses local time, the API should define how local booking times are interpreted and

stored. Database checks should enforce basic validity, while the application enforces facility-specific rules.

Reservation status should not substitute for payment status or physical occupancy. A reservation can be confirmed while payment is pending in a supported workflow, and a space can be physically occupied without a confirmed reservation. Keeping these concepts separate reduces ambiguity and supports later integration.[8]

4.5 Reservation and Availability Logic

For a requested interval [requested_start, requested_end), an existing active booking conflicts when its start is earlier than requested_end and its end is later than requested_start. Cancelled bookings are excluded from the active-conflict query. Pending reservations should be included only if the design gives them a temporary hold, in which case the hold must expire reliably.

The availability service validates that the requested interval is positive and falls within facility operating rules. It filters spaces by facility status, vehicle category, space status, and restrictions. For each eligible space, it checks active reservations for an overlapping interval. Results are returned to the frontend as a time-specific snapshot.

When a user requests confirmation, the reservation service repeats the conflict check inside a transaction and applies the selected database concurrency-control strategy. A simple check-then-insert without a transaction or suitable locking can allow simultaneous requests to pass the same check. The implementation must be tested with concurrent requests and should guarantee that conflicting confirmed reservations cannot both be committed.

Cancellation changes the reservation state and releases the interval for future availability according to policy. Check-in and completion should follow permitted state transitions. If the facility allows a grace period, early entry, or buffer between bookings, the conflict rule must account for those rules consistently.[7]

4.6 API Design

Table No.1 Rest API

POST /api/auth/register: Create a user account after validation.
POST /api/auth/login: Authenticate a user and establish the configured session or token.
GET /api/facilities: List facilities with supported search filters.
GET /api/facilities/{facilityId}: Retrieve facility details and public restrictions.
GET /api/facilities/{facilityId}/availability: Query eligible spaces for a time window and vehicle category.
POST /api/reservations: Create a reservation after server-side validation and conflict prevention.
GET /api/reservations/me: List reservations belonging to the authenticated user.
GET /api/reservations/{reservationId}: Retrieve a booking if the caller is authorized.
POST /api/reservations/{reservationId}/cancel: Cancel an eligible reservation under the configured policy.
GET /api/admin/facilities: List facilities within the administrator's authorized scope.
POST /api/admin/facilities: Create a facility record.
POST /api/admin/spaces: Create or configure a parking space.
PATCH /api/admin/reservations/{reservationId}/status: Update a permitted status with authorization and audit logging.
GET /api/admin/reports/summary: Return authorized aggregate booking and utilization summaries.

API responses should use appropriate HTTP status codes and structured error bodies. A conflicting reservation may return a conflict response with a clear error code, while invalid intervals or missing required fields should return validation errors. Sensitive stack traces, database details, password hashes, and internal information must not be returned to clients.

4.7 Security and Privacy Design

Authentication verifies identity, while authorization determines what the user may do. Every reservation read or update should verify ownership or administrative permission on the server. Hiding a button in the frontend is not an access-control mechanism. Administrative endpoints should require explicit roles and, where appropriate, facility-scoped permissions.

The application should use parameterized database access or a safe ORM, validate and normalize input, encode output appropriately, and protect against common web risks. Passwords should be stored using a modern password-hashing algorithm if local authentication is used. Tokens and sessions must be protected, and secrets kept in environment configuration or a secret manager rather than source code.

Personal details and vehicle identifiers should be visible only to users who need them. Logs should use booking identifiers where possible and avoid passwords, access tokens, payment credentials, or unnecessary personal data. HTTPS, database backups, retention limits, secure file handling, and incident-response procedures should be considered before public deployment.[4]

4.8 Deployment and Scalability

A small deployment can host the frontend, backend API, and relational database as separate components. Environment variables should distinguish development, test, and production settings. Database migrations should be version-controlled and applied in a controlled sequence. Backups and restore procedures should be tested.

Indexes on facility identifiers, space identifiers, reservation status, and time fields can improve search and conflict-check performance. Stable facility details may be cached, but live availability and reservation confirmation must account for current committed bookings. If the system scales to multiple application instances, concurrency correctness must be guaranteed by shared transactional storage rather than an in-memory lock local to one server.

Monitoring should track request latency, error rates, failed bookings, conflict rejections, notification failures, and database health. If map or payment services are integrated, timeouts and retries should be bounded and external failures should not create duplicate reservations or charges. Idempotency keys may be useful for retryable requests.[2]

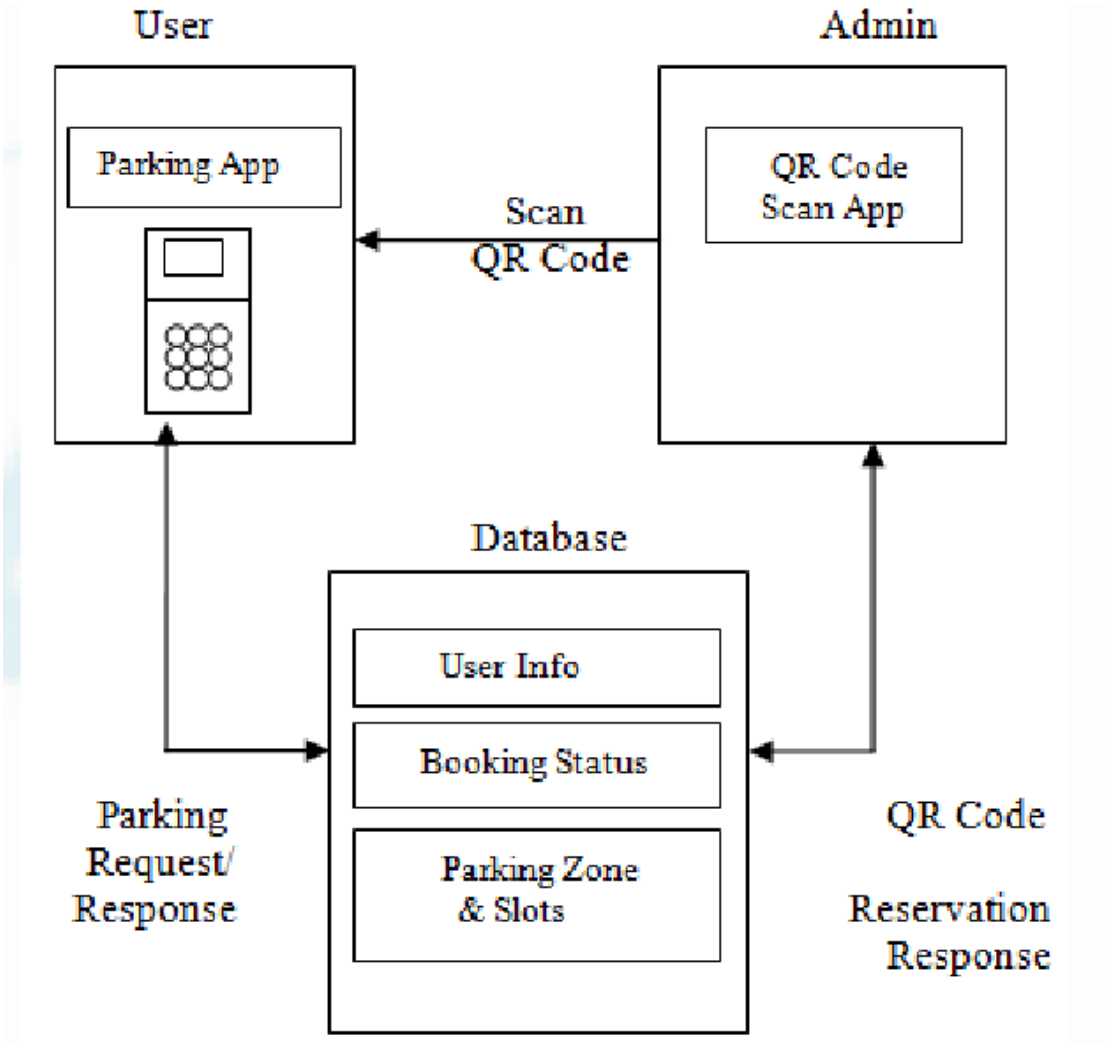


Fig 4.2 Deployment Architecture

CHAPTER-5
IMPLEMENTATION AND TESTING

5.1 Input Design

The input design should make the reservation process straightforward while collecting only information needed for the requested operation. Public search should allow the user to choose a facility or search area, arrival time, departure time, and vehicle category. Search results should show rates, restrictions, operating hours, and the time window to which availability applies.

A reservation form should display the selected facility and space category, start and end time, price estimate, cancellation terms, and any vehicle details required by the facility. Before submission, the user should review the request. The server must recalculate relevant values and validate all fields rather than trusting client-submitted prices or availability.[8]

Input Fields and Validation Rules

- Facility or search area must correspond to supported facility or valid search criteria.
- Start and end times are required; end time must be later and comply with facility rules.
- Vehicle category is required where facilities or spaces have category restrictions.
- Space selection must refer to an active eligible space in the selected facility.
- Reservation creation and management require authentication.
- Contact information should be validated and collected only when needed.
- If payment is enabled, use a provider's secure hosted or tokenized flow rather than collecting card data directly.
- Cancellation must reference a reservation owned by the user or accessible to an authorized administrator.
- Administrative fields must be validated for rates, effective dates, and authorized facility scope.

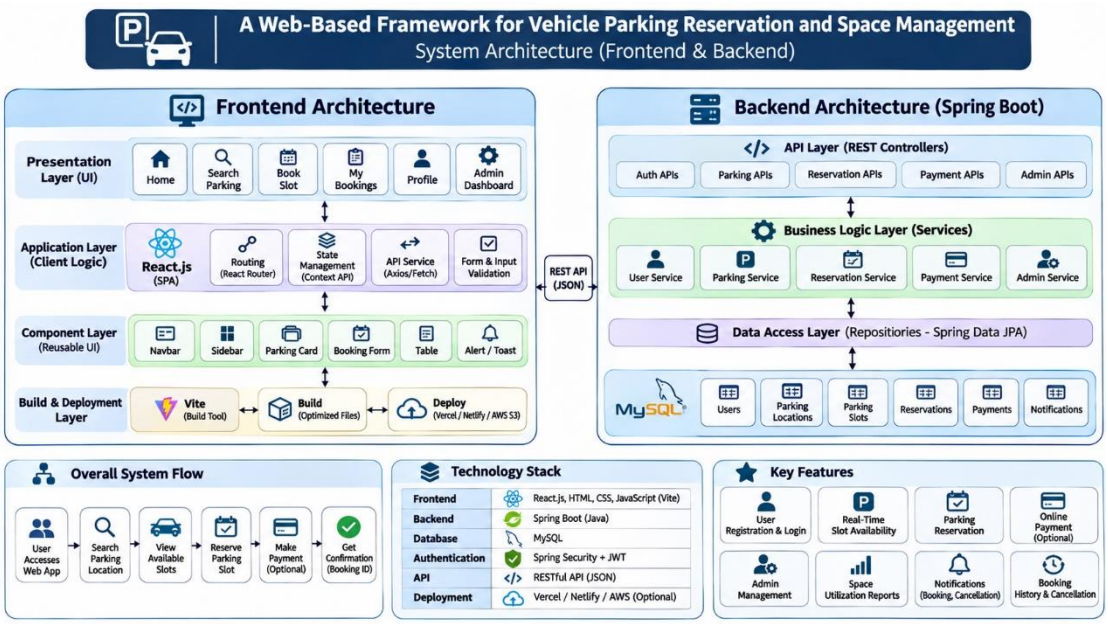


Fig 5.1 Frontend Architecture

5.2 Output Design

The facility listing should show the facility name, address, opening hours, supported vehicle categories, rate information, and relevant restrictions. Search results should distinguish available reservation options from general facility capacity. If the system does not use sensors, the interface should avoid describing a space as physically vacant in real time.

A booking confirmation should include a unique reference, facility and space details, start and end time with time zone, amount or pricing basis, booking status, cancellation terms, and arrival instructions. Booking history should show upcoming and past reservations with status labels. Cancellation feedback should state whether the booking was cancelled and whether any refund is pending if payment integration exists.

The administrative dashboard can summarize reservations, cancellations, check-ins, no-shows, and estimated utilization for a selected period. Utilization calculations should state their denominator and assumptions; reservation counts alone do not prove physical occupancy. Reports should allow authorized users to filter by facility and date and export only the data needed.[8]

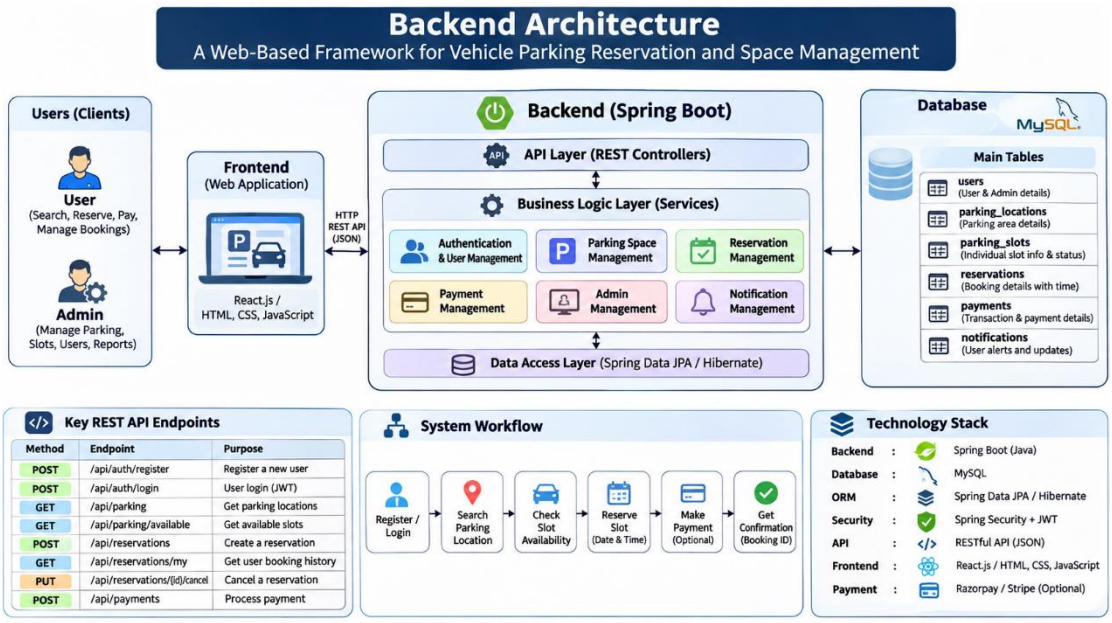


Fig 5.2 Backend Architecture

5.3 Implementation Approach

The application can use a modern JavaScript frontend such as React or Angular, a backend API such as Spring Boot, and a relational database such as PostgreSQL or MySQL. The final selection should match the development team's skills and deployment environment. The key requirement is that reservation rules live in a server-side service and are enforced consistently for every client.

Implementation should begin with the data model and user roles. The next phase should provide facility and space administration, followed by search and availability calculation. Reservation creation, cancellation, and status transitions should then be implemented with transactional conflict prevention. After the core flow works, the project can add notifications, reports, optional payment handling, and deployment configuration.

The frontend should use reusable components for search forms, facility cards, reservation forms, booking history, status indicators, and administrator tables. The backend should separate controllers, request/response DTOs, validation, services, repositories, and security configuration. Database access should use parameterized queries or a safe ORM. API contracts should be documented so frontend and backend changes can be tested together.

Reservation creation and payment processing should be treated as separate but coordinated operations if online payment is enabled. A payment provider may confirm

asynchronously, so the application needs a defined state machine and idempotent handling of callbacks. A successful booking notification must not be sent before the reservation state is durably committed.[9]

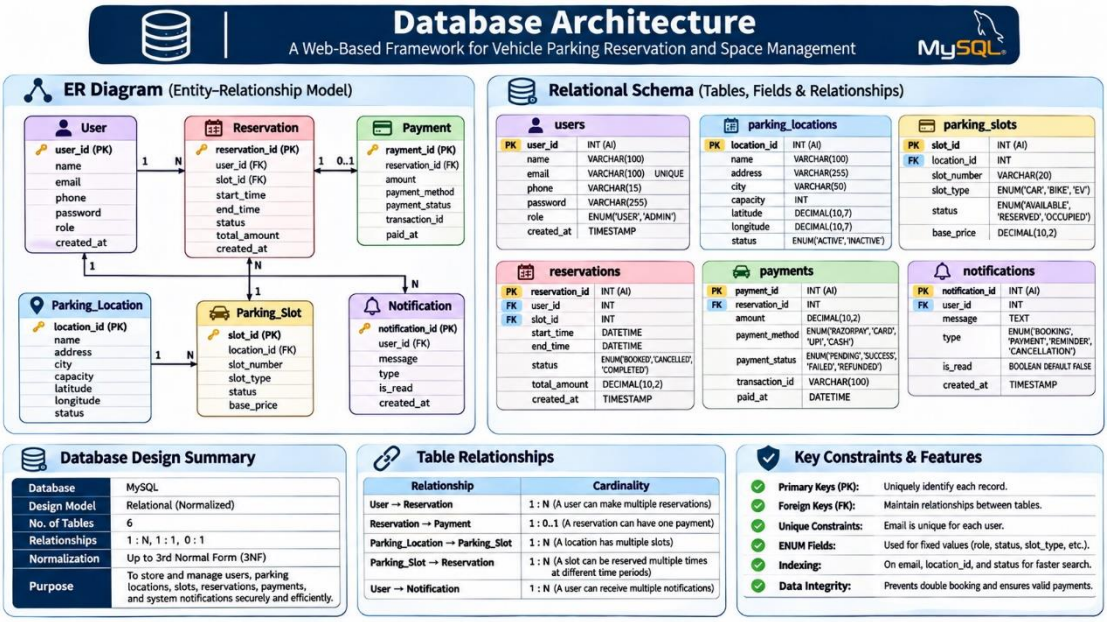


Fig 5.3 Database Architecture

5.4 Validation and Error Handling

Validation errors should explain which field or rule failed. Examples include an end time before the start time, a facility that is closed, an inactive space, an unsupported vehicle category, an expired session, a booking outside the cancellation window, or a conflicting reservation. The interface should preserve appropriate form data after recoverable errors.

Conflict errors are expected outcomes in a concurrent reservation system and should not be treated as unexpected server failures. If another user reserves the space first, the application should report that it is no longer available and refresh options. Database failures should trigger transaction rollback, and the user should receive a safe message without internal details.

Notifications may fail after a reservation has been saved. The application should retain the confirmed booking and record notification failure for retry or review rather than deleting the booking automatically. A client retry after timeout should not create duplicate reservations; idempotency controls or safe duplicate-request handling should be considered.[10]

5.5 Testing Strategy

Unit tests should cover interval-overlap logic, pricing calculation, cancellation eligibility, status transitions, facility operating hours, and authorization decisions. Repository or integration tests should verify persistence, foreign-key relationships, transaction rollback, and conflict prevention. API tests should cover successful requests, malformed input, unauthenticated access, unauthorized access, and expected conflict responses.

End-to-end tests should exercise a complete user journey: search a facility, inspect availability, reserve a space, view the booking, cancel it, and confirm that the interval becomes available again when policy permits. Administrative tests should cover creating and disabling spaces, updating rates, viewing reports, and preventing unauthorized changes.

Concurrency testing is essential. The expected result is that no more than one conflicting reservation is confirmed. Additional tests should verify that adjacent non-overlapping intervals are accepted when the configured turnover rule allows them, and cancelled bookings no longer block availability.

Performance tests should use a documented dataset and workload. Measurements may include search latency, reservation latency, throughput, error rate, and database load. Report the environment and number of simulated users rather than claiming the system is universally fast. Security testing should review access control, input handling, session protection, and sensitive-data exposure.[9]

5.6 Test Cases

Table No.2 Test Cases

Test ID	**Scenario**	**Expected Result**	Expected Result
TC-01	Browse facilities	Request facility list with valid filters.	Eligible facility records and public details are returned.
TC-02	Invalid time interval	Submit an end time earlier than or equal to the start time.	Request is rejected with a validation message.
TC-03	Valid reservation	Reserve an eligible available space for a valid interval.	One confirmed reservation and unique reference are returned.

TC-04	Overlapping booking	Reserve a space whose interval overlaps an active reservation.	The conflicting reservation is rejected.
TC-05	Adjacent bookings	Create consecutive intervals with a shared boundary where policy permits.	Both bookings are accepted without overlap.
TC-06	Concurrent requests	Submit simultaneous overlapping requests for the same space.	At most one conflicting booking is confirmed.
TC-07	Cancelled booking	Cancel an eligible reservation and search the interval again.	The cancelled interval is available subject to policy.
TC-08	Unauthorized booking access	Attempt to read another user's reservation.	Access is denied without revealing private details.
TC-09	Administrator authorization	Call a management endpoint as a regular user.	The request is rejected.
TC-10	Inactive space	Attempt to reserve a disabled space.	The request is rejected with a useful message.
TC-11	Facility closed	Request a time outside permitted operating hours.	Booking is rejected or handled according to configured policy.
TC-12	Notification failure	Simulate a provider failure after booking commit.	Reservation remains consistent and notification failure is recorded.
TC-13	Repeated request	Retry a booking request after a simulated timeout.	Duplicate bookings are prevented according to idempotency rules.
TC-14	Report accuracy	Create known reservations and request a summary.	Counts match stored records and stated calculation rules.

CHAPTER-6
SOFTWARE ENVIRONMENT AND FUTURE ENHANCEMENTS

6.1 Software Requirements

Table No.3 Software Requirements

Software	Purpose
Operating System	Windows, Linux, or macOS
Java 17 or 21	Backend runtime and development
Spring Boot	REST API development
Maven	Build and dependency management
React	Frontend development
Node.js and npm	Frontend build tooling
PostgreSQL or MySQL	Persistent storage
Docker Engine	Container image creation and execution
Kubernetes	Container orchestration
kubectl	Kubernetes cluster management
Git	Version control
Postman	API testing
IDE	IntelliJ IDEA, Eclipse, or VS Code

The exact Spring Boot version and Java version should be selected from a compatible supported combination.

6.2 Hardware Requirements

The following are suggested development requirements, not mandatory production specifications.

Table No.4 Hardware Requirements

Component	Suggested Requirement
Processor	Modern multi-core processor
RAM	8 GB minimum; 16 GB or more recommended for local Kubernetes
Storage	20 GB or more available space
Network	Internet connectivity for dependency downloads and image retrieval
Client device	Desktop, laptop, tablet, or smartphone

Production infrastructure should be sized according to expected users, concurrent sessions, media delivery needs, and database workload.

6.3 Future Enhancements

1. Real-Time Parking Slot Detection

The system can be integrated with IoT sensors or cameras to detect whether parking spaces are occupied or vacant in real time.

Example: When a vehicle leaves a parking slot, the system automatically updates its status to *Available*.

Benefit: Reduces the time users spend searching for parking spaces.

2. Mobile Application Development

A dedicated Android and iOS application can be developed alongside the web application.

Users could:

- Find nearby parking locations.
- Reserve slots using their smartphones.
- Receive booking confirmations and reminders.
- View their current and previous bookings.

Benefit: Makes the system more accessible and convenient.

3. GPS-Based Parking Location Finder

GPS and digital maps can be integrated to help users locate parking facilities near their current location.

The system could display:

- Nearby parking areas.

- Distance to each parking facility.
- Available parking spaces.
- Directions to the selected location.

Benefit: Helps drivers find suitable parking facilities more quickly.

4. Online Payment Integration

The system can support online payments through UPI, debit cards, credit cards, and other payment gateways.

Additional features could include:

- Automatic parking fee calculation.
- Digital receipts.
- Refund processing for eligible cancellations.
- Online payment history.

Benefit: Reduces manual payment handling and improves transaction convenience.

5. Automated Entry and Exit Management

The system could integrate QR codes, RFID cards, or number-plate recognition to automate vehicle entry and exit.

Example: After a user reserves a slot, the system generates a QR code. The parking attendant scans it to verify the reservation.

Benefit: Reduces manual verification and speeds up vehicle entry.

6. Dynamic Parking Pricing

Parking fees could be adjusted according to demand, duration, location, and time of day.

Example: Parking during peak business hours might cost more than parking during less busy hours.

Benefit: Helps parking operators manage demand and potentially improve revenue.

Note: Pricing rules should be transparent and comply with applicable regulations.

7. AI-Based Parking Demand Prediction

Machine learning models could analyse historical booking records, peak hours, and parking usage patterns to estimate future demand.

Example: The system might predict that a particular parking facility will have high demand on weekday mornings.

Benefit: Helps parking operators plan capacity and helps users make better parking decisions.

8. Automatic Booking Expiry and Slot Release

If a user fails to arrive within the permitted reservation period, the system could automatically expire the booking and release the slot.

The system could also:

- Send reminders before a reservation expires.
- Notify users about cancellations.
- Make released slots available to other users.

Benefit: Prevents parking spaces from remaining unnecessarily reserved.

9. Electric Vehicle Charging Station Integration

The system could allow electric vehicle owners to find and reserve parking spaces equipped with charging stations.

Possible features include:

- Identifying charging-enabled parking slots.
- Checking charger availability.
- Reserving a charging slot.
- Displaying charging costs and session details.

Benefit: Makes the platform more useful as electric vehicle adoption increases.

10. Advanced Security and Fraud Prevention

The system could introduce stronger security measures, including:

- Multi-factor authentication.
- Secure online payments.
- Role-based access control for administrators and parking operators.
- Detection of duplicate or suspicious reservations.
- Audit logs for important activities.

Benefit: Protects user information and reduces unauthorized access and fraudulent bookings.[10]

CONCLUSION

The Web-Based Framework for Vehicle Parking Reservation and Space Management provides an effective solution to the common problems associated with finding and managing parking spaces. The system enables users to search for parking facilities, check slot availability, reserve parking spaces in advance, and manage their bookings through a web-based platform. It also helps administrators manage parking slots, monitor reservations, and maintain parking-related records efficiently.

By digitizing the parking reservation process, the proposed framework can reduce the time spent searching for parking, minimize booking conflicts, improve space utilization, and enhance the overall user experience. It also provides a centralized platform for managing parking operations, reducing the need for manual record-keeping and improving the accuracy of reservation information.

In conclusion, this project demonstrates how web technologies can be applied to address practical parking management challenges through a convenient, accessible, and organized system. In the future, the framework can be enhanced with real-time parking detection using IoT sensors, GPS-based navigation, online payment integration, QR-code-based vehicle verification, and AI-based parking demand prediction. These improvements could transform the proposed framework into a more intelligent and scalable parking management solution for smart cities.[10]

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